

Frozen provider components

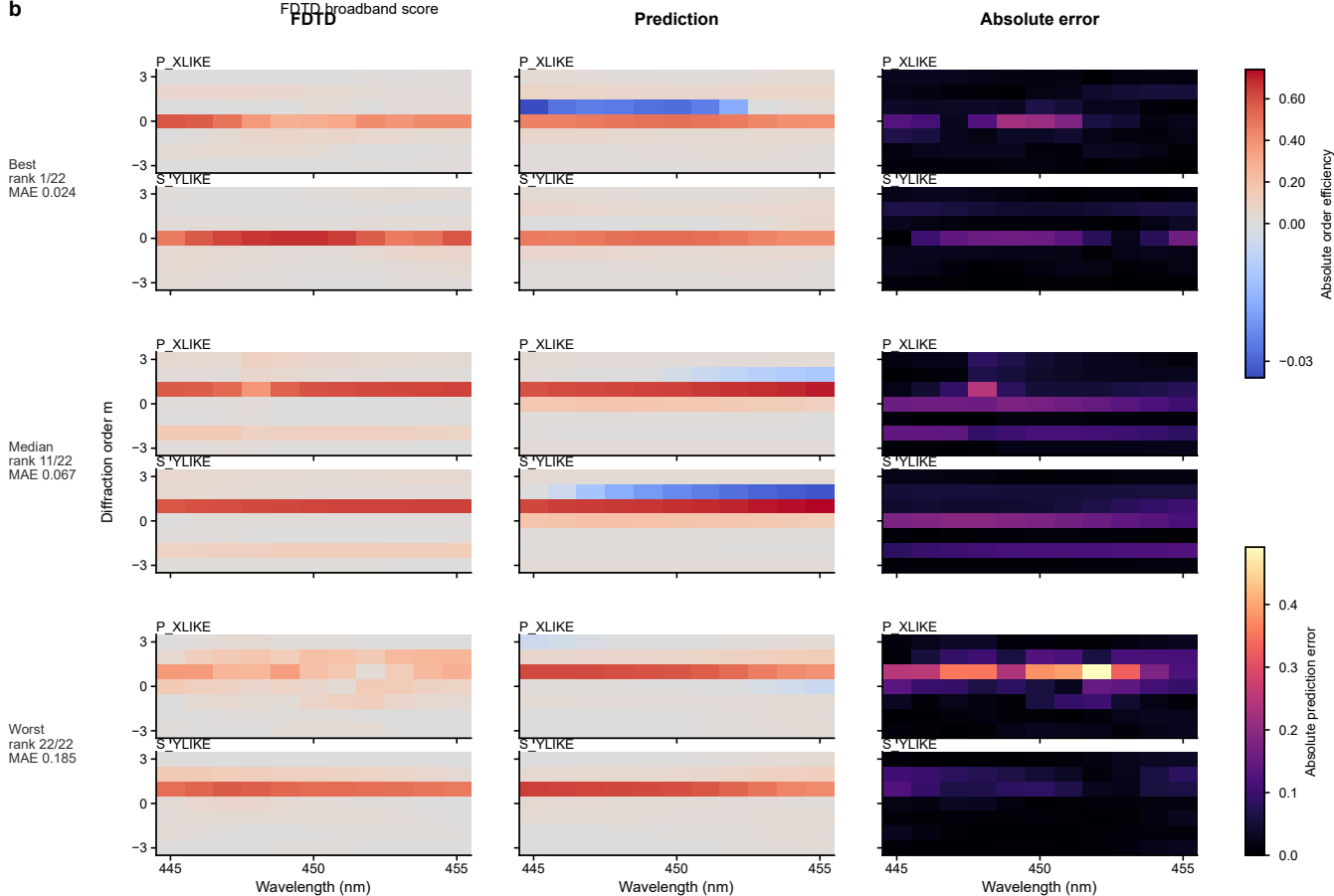
Ranking LF_only / ensemble_raw

Spectra LF_ridge_residual / ensemble_raw

Distinct components; not a universal surrogate

$u_x = 0$, $k_y = 0$ · explicit P/S · 445–455 nm

b



Screening/ranking only; final quantitative verification uses FDTD.